

EXECUTIVE SUMMARY · GALATIQ CASE STUDY

Invoice automation for Acme Corp

A multi-agent accounts-payable pipeline, reasoning on Grok, that reads, checks, approves and pays invoices in seconds.

Verified on live Grok (grok-4.7) with the case's 20 sample invoices · 2 October 2026

BUSINESS IMPACT

Most invoices now clear without a person, in seconds rather than days

80%

of invoices decided
without a human

Today every invoice is keyed
by hand

34 s

median time
per invoice

Against a 5-day
manual cycle

\$204K

of bad invoices
blocked

Stopped before any
money moved

0

invoices paid twice

All 3 duplicates held
for review

Baseline from the brief: about \$2M a year in manual processing cost, 30% data-entry errors, 5-day cycles.

BUSINESS IMPACT

Every planted problem was stopped before money moved

3

Duplicate payments prevented

The same invoice sent as PDF and text, plus a revised copy. Each was held, with a note of exactly what changed.

1

Fraudulent invoice stopped

\$100,000 from "Fraudster LLC": a zero-stock item, "URGENT" pressure, a wire-transfer demand and a due date of "yesterday".

4

Arithmetic errors caught

Including a \$50 overcharge hidden in a \$22,562.80 grand total, the kind of slip manual keying lets through.

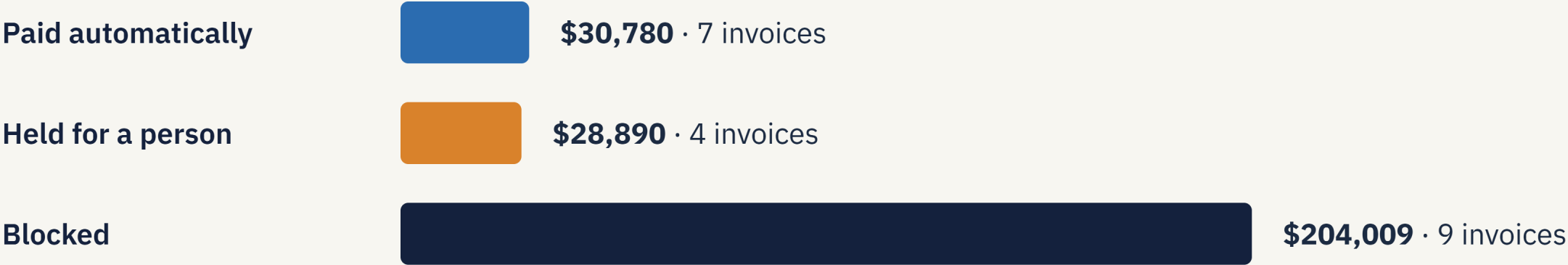
7

Stock and catalog problems

5 invoices billed more than is in stock (counted across repeated lines) and 2 billed items we don't carry.

BUSINESS IMPACT

Where the money went



Held invoices arrive pre-analysed: 3 duplicates, plus 1 invoice Grok judged risky enough to check (a vendor rename, \$25 under the \$10K approval threshold).

Blocked amounts are exposure avoided, not savings. Most of it is one \$100,000 fraudulent invoice and four invoices billing for stock we don't have.

BENEFITS

People handle the exceptions, with the reasoning already done

GROK'S NOTE ON INVOICE 1012

"Two risk flags together still warrant a hold. ... A reviewer should verify the legal name change and payee bank details, and confirm the amount and PO-20260115 were not structured to stay under VP scrutiny."

-  Every decision explained in plain language
-  A full audit trail, step by step, for every invoice
-  The same approval policy, applied every time
-  A review queue with the evidence gathered

ASSUMPTIONS

What we assumed where the brief was silent

Assumption	Why
Only Grok is real; bank, ERP and FX are simulated	Brief: call Grok, simulate everything else
Paying an invoice doesn't reduce stock	Same result whatever the processing order
Duplicates go to a person, never netted	A claimed PO amendment can't be verified
Foreign currency converts at a fixed rate	Keeps the \$10K VP rule in dollars
Prices over 10% above catalog are flagged only	Catalog prices are our own addition
grok-4.7 instead of the brief's grok-3	xAI no longer offers grok-3
Impact shown as rates, not dollar savings	The brief gives no invoice volume

ASSUMPTIONS

Design principle: the AI judges, the rules decide



Rules can't be loosened

Stock, arithmetic, duplicate and fraud checks are code. Grok can escalate a decision, but never approve what the policy rejects.



Only code moves money

Payment is a fixed step after approval, with its own checks. Grok has no way to trigger a payment.



No check is left to memory

If Grok skips a check, a guard runs it. If Grok is unavailable, structured invoices still get a policy decision.

Seen live: Grok held invoice 1012, which policy alone would have paid. The stricter outcome stood, and the disagreement was logged.

CRITERIA FULFILLED

The brief's four-stage workflow, end to end

1

Ingestion

Reads text, email, PDF, JSON, CSV and XML, and repairs OCR-style errors like "2026".

2

Validation

Checks stock, arithmetic, catalog price, duplicates, currency and fraud signals.

3

Approval

Policy plus Grok's judgement, critiqued before it stands. Over \$10K gets VP-level scrutiny.

4

Payment

Pays approved invoices through the provided mock bank and records why others stopped.

✓ Grok reasoning

✓ LangGraph agents

✓ Tool calling

✓ Structured output

✓ Self-correction

✓ SQLite inventory

✓ Runs locally

CRITERIA FULFILLED

All seven evaluation criteria are covered

Criterion	How it is met
Functionality	All 20 invoices processed; 19 of 20 match expected on live Grok
Code architecture	LangGraph, typed contracts, 99 offline tests, per-step audit log
Agent sophistication	Tool calling, structured output, self-correction, a critique loop
Shipping mentality	Scoped MVP; runs offline with no key; extras deferred on purpose
Business comms	Impact summary with every batch; plain-language rationales
Innovation	Duplicate ledger, fraud signals, catalog price and currency checks
User experience	A readable panel per invoice, a review queue, JSON for systems

NEXT STEPS

Taking it to production

- **Pilot on a month of real invoices** to size the share of the \$2M it removes
- **Connect the vendor master and purchase orders** so renames and revisions become hard checks
- **Add a review screen** that resumes the pipeline once a person decides
- **Process invoices in parallel** to cut batch time (about 34 seconds each today)
- **Live exchange rates and OCR** for scanned, image-only PDFs